UNIT II:-Software Quality Assurance

Software Quality Assurance:-

Software Quality Assurance (SQA) is simply a way to assure quality in the software. It is the set of activities that ensure processes, procedures as well as standards are suitable for the project and implemented correctly. It is a process that works parallel to Software Development. It focuses on improving the process of development of software so that problems can be prevented before they become major issues. Software Quality Assurance is a kind of Umbrella activity that is applied throughout the software process.

For those looking to deepen their expertise in SQA and elevate their professional skills, consider exploring a specialized training program **Manual to Automation Testing: A QA Engineer's Guide** . This program offers practical, hands-on experience and advanced knowledge that complements the concepts covered in this guide.

Quality

Quality in a product or service can be defined by several measurable characteristics. Each of these characteristics plays a crucial role in determining the overall quality.

Generally, the quality of the software is verified by third-party organizations like international standard organizations .

Elements of Software Quality Assurance (SQA)

- **Standards:** The IEEE, ISO, and other standards organizations have produced a broad array of software engineering standards and related documents. The job of SQA is to ensure that standards that have been adopted are followed and that all work products conform to them.
- **Reviews and audits:** Technical reviews are a quality control activity performed by software engineers for software engineers. Their

intent is to uncover errors. Audits are a type of review performed by SQA personnel (people employed in an organization) with the intent of ensuring that quality guidelines are being followed for software engineering work.

- **Testing:** Software testing is a quality control function that has one primary goal to find errors. The job of SQA is to ensure that testing is properly planned and efficiently conducted for primary goal of software.
- **Error/defect collection and analysis** : SQA collects and analyzes error and defect data to better understand how errors are introduced and what software engineering activities are best suited to eliminating them.
- **Change management:** SQA ensures that adequate change management practices have been instituted.
- **Education:** Every software organization wants to improve its software engineering practices. A key contributor to improvement is education of software engineers, their managers, and other stakeholders. The SQA organization takes the lead in software process improvement which is key proponent and sponsor of educational programs.
- **Security management:** SQA ensures that appropriate process and technology are used to achieve software security.
- **Safety:** SQA may be responsible for assessing the impact of software failure and for initiating those steps required to reduce risk.
- **Risk management** : The SQA organization ensures that risk management activities are properly conducted and that risk-related contingency plans have been established.

Focus of Software Quality Assurance (SQA)

The Software Quality

Assurance (SQA) focuses on the following :

Software Quality Assurance (SQA)

- **Software's Portability:** Software's portability refers to its ability to be easily transferred or adapted to different environments or platforms without needing significant modifications. This ensures that the software can run efficiently across various systems, enhancing its accessibility and flexibility.
- **Software's Usability:** Usability of software refers to how easy and intuitive it is for users to interact with and navigate through the application. A high level of usability ensures that users can effectively accomplish their tasks with minimal confusion or frustration, leading to a positive user experience.

- **Software's Reusability:** Reusability in software development involves designing components or modules that can be reused in multiple parts of the software or in different projects. This promotes efficiency and reduces development time by eliminating the need to reinvent the wheel for similar functionalities, enhancing productivity and maintainability.
- **Software's Correctness:** Correctness of software refers to its ability to produce the desired results under specific conditions or inputs. Correct software behaves as expected without errors or unexpected behaviors, meeting the requirements and specifications defined for its functionality.
- **Software's Maintainability:** Maintainability of software refers to how easily it can be modified, updated, or extended over time. Well maintained software is structured and documented in a way that allows developers to make changes efficiently without introducing errors or compromising its stability.
- **Software's Error Control:** Error control in software involves implementing mechanisms to detect, handle, and recover from errors or unexpected situations gracefully. Effective error control ensures that the software remains robust and reliable, minimizing disruptions to users and providing a smoother experience overall.

Major Activities in Software Quality Assurance (SQA)

- **SQA Management Plan:** Make a plan for how you will carry out the SQA throughout the project. Think about which set of software engineering activities are the best for project. Check level of SQA team skills.
- **Set The Check Points:** SQA team should set checkpoints. Evaluate the performance of the project on the basis of collected data on different check points.
- **Measure Change Impact:** The changes for making the correction of an error sometimes re introduces more errors keep the measure of

impact of change on project. Reset the new change to check the compatibility of this fix with whole project.

- **Multi testing Strategy:** Do not depend on a single testing approach. When you have a lot of testing approaches available use them.
- **Manage Good Relations:** In the working environment managing good relations with other teams involved in the project development is mandatory. Bad relation of SQA team with programmer's team will impact directly and badly on project.
- **Maintaining records and reports:** Comprehensively document and share all QA records, including test cases, defects, changes, and cycles, for stakeholder awareness and future reference.
- **Reviews software engineering activities:** The SQA group identifies and documents the processes. The group also verifies the correctness of software product.
- **Formalize deviation handling:** Track and document software deviations meticulously. Follows established procedures for handling variances.

Benefits of Software Quality Assurance (SQA)

- SQA produces high quality software.
- High quality application saves time and cost.
- SQA is beneficial for better reliability.
- SQA is beneficial in the condition of no maintenance for a long time.
- High quality commercial software increase market share of company.
- Improving the process of creating software.
- Improves the quality of the software.

Software Review and Its types

Software Review is a systematic inspection of software by one or more individuals who work together to find and resolve errors and defects in the software during the early stages of the Software Development Life Cycle (SDLC). A software review is an essential part of the Software Development Life Cycle (SDLC) that helps software engineers in validating the quality, functionality, and other vital features and components of the software. It is a whole process that includes testing the software product and it makes sure that it meets the requirements stated by the client.

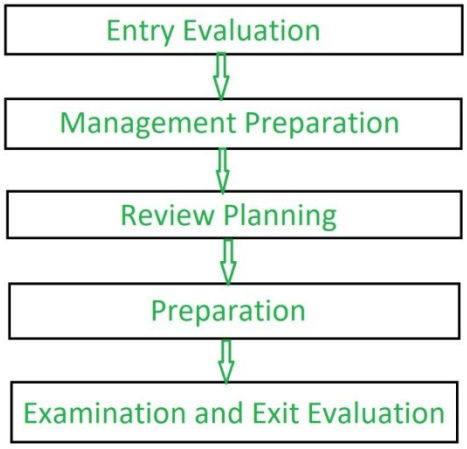
Usually performed manually, software review is used to verify various documents like requirements, system designs, codes, test plans, and test cases.

Objectives of Software Review

The objective of the software review is:

1. To improve the productivity of the development team.
2. To make the testing process time and cost-effective.
3. To make the final software with fewer defects.
4. To eliminate the inadequacies.

Process of Software Review



1. **Entry Evaluation:** By confirming documentation, fulfilling entry requirements and assessing stakeholder and team preparation, you can determine the software's availability.
2. **Management Preparation**: To get ready for the review process, assign roles, gather resources and provide brief management.
3. **Review Planning:** Establish the review's goals and scope, invite relevant parties and set a time for the meeting.
4. **Preparation**: Distribute appropriate resources, give reviewers time to get familiar and promote issue identification to help them prepare.
5. **Examination and Exit Evaluation:** Reviewers should collaborate to examine the results, record concerns, and encourage candid communication in meetings. It assess the results, make remedial plans based on flaws that have been reported and assess the process's overall efficacy.

Types of Software Reviews

There are mainly 3 types of software reviews:

1. Software Peer Review

Peer review is the process of assessing the technical content and quality of the product and it is usually conducted by the author of the work product along with some other developers.

Peer review is performed in order to examine or resolve the defects in the software, whose quality is also checked by other members of the team.

Peer Review has following types:

1. **Code Review:** Computer source code is examined in a systematic way.
2. **Pair Programming:** It is a code review where two developers develop code together at the same platform.
3. **Walkthrough:** Members of the development team is guided by author and other interested parties and the participants ask questions and make comments about defects.
4. **Technical Review:** A team of highly qualified individuals examines the software product for its client's use and identifies technical defects from specifications and standards.
5. **Inspection:** In inspection the reviewers follow a well-defined process to find defects.

2. Software Management Review

Software Management Review evaluates the work status. In this section decisions regarding downstream activities are taken.

3. Software Audit Review

Software Audit Review is a type of external review in which one or more critics, who are not a part of the development team, organize an independent inspection of the software product and its processes to assess their compliance with stated specifications and standards. This is done by managerial level people.

Advantages of Software Review

1. Defects can be identified earlier stage of development (especially in formal review).

2. Earlier inspection also reduces the maintenance cost of software.
3. It can be used to train technical authors.
4. It can be used to remove process inadequacies that encourage defects.

FORMAL TECHNICAL REVIEWS:-

A formal technical review is a software quality assurance activity performed by software engineers (and others). The objectives of the FTR are

(1) To uncover errors in function, logic, or implementation for any representation of the software;

(2) To verify that the software under review meets its requirements;

(3) To ensure that the software has been represented according to predefined standards;

(4) To achieve software that is developed in a uniform manner; and

(5) To make projects more manageable. In addition, the FTR serves as a training ground, enabling junior engineers to observe different approaches to software analysis, design, and implementation. The FTR also serves to promote backup and continuity because a number of people become familiar with parts of the software that they may not have otherwise seen

1)**The Review Meeting:-**

Regardless of the FTR format that is chosen, every review meeting should abide by the following constraints:

• Between three and five people (typically) should be involved in the review.

• Advance preparation should occur but should require no more than two hours of work for each person.

• The duration of the review meeting should be less than two hours.

Given these constraints, it should be obvious that an FTR focuses on a specific (and small) part of the overall software. For example, rather than attempting to review an entire design, walkthroughs are conducted for each component or small group of components. By narrowing focus, the FTR has a higher likelihood of uncovering errors.

At the end of the review, all attendees of the FTR must decide whether to

(1) Accept the product without further modification,

(2) Reject the product due to severe errors (once corrected, another review must be performed), or

(3) Accept the product provisionally (minor errors have been encountered and must be corrected, but no additional review will be required). The decision made, all FTR attendees complete a sign-off, indicating their participation in the review and their concurrence with the review team's finding

2) Review Reporting and Record Keeping:

During the FTR, a reviewer (the recorder) actively records all issues that have been raised. These are summarized at the end of the review meeting and a review issues list is produced. In addition, a formal

technical review summary report is completed. A review summary report answers three questions:

1. What was reviewed?

2. Who reviewed it?

3. What were the findings and conclusions?

The review summary report is a single page form (with possible attachments). It becomes part of the project historical record and may be distributed to the project leader and other interested parties. The review issues list serves two purposes:

(1) to identify problem areas within the product and

(2) to serve as an action item checklist that guides the producer as corrections are made.

An issues list is normally attached to the summary report

3)Review Guidelines

Guidelines for the conduct of formal technical reviews must be established in advance, distributed to all reviewers, agreed upon, and then followed

1)Review the product, not the producer. An FTR involves people and egos. Conducted properly, the FTR should leave all participants with a warm feeling of accomplishment

2) Set an agenda and maintain it. One of the key maladies of meetings of all types is drift. An FTR must be kept on track and on schedule

3)Limit debate and rebuttal. When an issue is raised by a reviewer, there may not be universal agreement on its impact

4)Enunciate problem areas, but don't attempt to solve every problem noted. A review is not a problem-solving session

5) Take written notes. It is sometimes a good idea for the recorder to make notes on a wall board, so that wording and priorities can be assessed by other reviewers as information is recorded.

6) Limit the number of participants and insist upon advance preparation. Two heads are better than one, but 14 are not necessarily better than 4 keep the number of people involved to the necessary minimum.

7) Develop a checklist for each product that is likely to be reviewed. A checklist helps the review leader to structure the FTR meeting and helps each reviewer to focus on important issues.

SOFTWARE RELIABILITY:-

There is no doubt that the reliability of a computer program is an important element of its overall quality. If a program repeatedly and frequently fails to perform, it matters little whether other software quality factors are acceptable. Software reliability is defined in statistical terms as "the probability of failure-free operation of a computer program in a specified environment for a specified time"

Measures of Reliability and Availability

Early work in software reliability attempted to extrapolate the mathematics of hardware reliability theory to the prediction of software reliability. Most hardware-related reliability models are

predicated on failure due to wear rather than failure due to design defects. In hardware, failures due to physical wear (e.g., the effects of temperature, corrosion, shock) are more likely than a design-related failure. Unfortunately, the opposite is true for software. In fact, all software failures can be traced to design or implementation problems; wear not enter into the picture. There has been debate over the relationship between key concepts in hardware reliability and their applicability to software Although an irrefutable link has yet be be established, it is worthwhile to consider a few simple concepts that apply to both system elements.

Software Safety

When software is used as part of the control system, complexity can increase by an order of magnitude or more. Subtle design faults induced by human error—something that can be uncovered and eliminated in hardware-based conventional control— become much more difficult to uncover when software is used. Software safety is a software quality assurance activity that focuses on the identification and assessment of potential hazards that may affect software negatively and cause an entire system to fail. If hazards can be identified early in the software engineering process, software design features can be specified that will either eliminate or control potential hazards. A modeling and analysis process is conducted as part of software safety. Initially, hazards are identified and categorized by criticality and risk. For example, some of the hazards associated with a computer-based cruise control for an automobile might be

• causes uncontrolled acceleration that cannot be stopped

• does not respond to depression of brake pedal (by turning off)

does not engage when switch is activated

- Slowly loses or gains speed

Software Quality Assurance Plan

The SQA Plan provides a road map for instituting software quality assurance. Developed by the SQA group, the plan serves as a template for SQA activities that are instituted for each software project. A standard for SQA plans has been recommended by the IEEE. Initial sections describe the purpose and scope of the document and indicate those software process activities that are covered by quality assurance. All documents noted in the SQA Plan are listed and all applicable standards are noted. The management section of the plan describes SQA's place in the organizational structure, SQA tasks and activities and their placement throughout the software process, and the organizational roles and responsibilities relative to product quality. The documentation section describes (by reference) each of the work products produced as part of the software process. These include

- project documents (e.g., project plan)
- models
- technical documents (e.g., specifications, test plans)
- user documents (e.g., help files)

In addition, this section defines the minimum set of work products that are acceptable to achieve high quality. The standards, practices, and conventions section lists all applicable standards and practices that are applied during the software process (e.g., document standards, coding standards, and review guidelines). In addition, all project, process, and

(in some instances) product metrics that are to be collected as part of software engineering work are listed. The reviews and audits section of the plan identifies the reviews and audits to be conducted by the software engineering team, the SQA group, and the customer. It provides an overview of the approach for each review and audit. The remainder of the SQA Plan identifies the tools and methods that support SQA activities and tasks; references software configuration management procedures for controlling change; defines a contract management approach; establishes methods for assembling, safeguarding, and maintaining all records; identifies training required to meet the needs of the plan; and defines methods for identifying, assessing, monitoring, and controlling risk.

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